Department of Computer Science

Developing Artificial Intelligence approaches to enhance representations of Turbulence in Atmospheric Models

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Introduction

Turbulence representation in atmospheric models is a critical challenge. Large Eddy Simulation (LES) models, while resolving large-scale eddies, model the sub-grid scale (SGS) components using parametrizations. Recent advances in Artificial Intelligence (AI) show promise in enhancing SGS representation, with neural networks achieving accuracy comparable to traditional models while improving computational efficiency (Pal, 2020; Vinuesa et al., 2022). This work aims to build on these advances by exploring AI-driven turbulence emulation in LES to enhance computational performance.

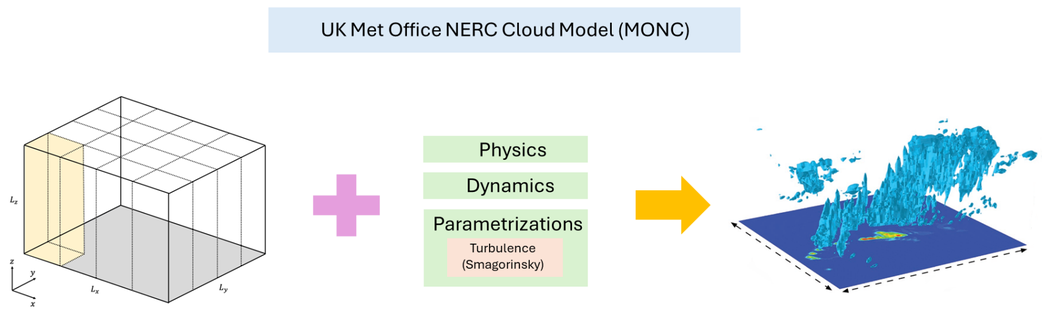


Fig 1: Schematic Representation of the LES setup

Objectives

- Explore AI based techniques, to learn turbulence closure models directly from high-resolution LES data, for reducing the **computational cost** of LES models.
- Conduct extensive evaluation/verification of the AI based **emulator** for various complex atmospheric turbulence settings.
- Utilize Explainable AI (**XAI**) methods to understand the behaviour of the AI models and to build trust.

Data and Methodology

- 3-D diagnostics data from the Met Office Natural Environment Research Council Cloud model (**MONC**).
- **Radiative Convective Equilibrium (RCE)** simulations.
- Diagnostic variables for training (**sampling freq = 1800 sec**).
  - **7 Input variables**: Wind velocity ($\boldsymbol{u}$,$\boldsymbol{v}$,$\boldsymbol{w}$), mixing ratios ($\boldsymbol{q}$), potential temperature ($\boldsymbol{\theta}$), Richardson number ($\boldsymbol{Ri_p}$)
  - **5 target variables**: Tendency fields for ($\boldsymbol{u}$,$\boldsymbol{v}$,$\boldsymbol{w}$), eddy viscosity ($\boldsymbol{\nu}$) and diffusivity ($\boldsymbol{\nu_h}$) coefficients
- Reduced 3-D data to ~5800 2-D slices across multiple time steps for training and evaluation using unseen MONC simulation data (**sampling frequency: 200 sec**).
- Four AI models chosen for the study: Convolutional Neural Network (**CNN**) with Gaussian error linear unit (**GELU**) and Rectified Linear Unit (**RELU**) activation functions; a U-shaped encoder-decoder network (**UNET**) and a simple version of Vision Transformer (**VIT**)

Results

Table 1: Summary of the 4 AI models trained in the proof-of-concept study.

Model	Layers	Activation function	Optimizer	Batch Size	Epochs	Learning Rate	Total no. of Params
CNN GELU	6 conv layers	GELU	ADAM	256	2000	0.001	~ 173 k
CNN RELU	6 conv layers	RELU	ADAM	256	2000	0.001	~ 173 k
VIT	Encoder Layer with 7 head attention	RELU	ADAM	64	2000	0.001	~ 0.5 B
UNET	5 conv layers, 1 fully connected layer	RELU	ADAM	256	2000	0.001	~ 581 k

❖ **Training platform**: **JASMIN GPU Cluster** with 1 **NVIDIA A100** GPU.

❖ **Training Times**: CNN GELU/RELU (~1.5 hours), VIT (~8 hours), UNET (~2.5 hours)

Figure 2 presents sample results where 7 input variables are used to predict 5 target fields, using 4 AI models (described in Table 1). The evaluation is based on two criteria:

- Data Structural Similarity Index (**DSSIM**) scores, evaluating structural similarity (Baker et al. 2023)
- **R2** scores, showing comparison of magnitudes of the targets vs predictions.

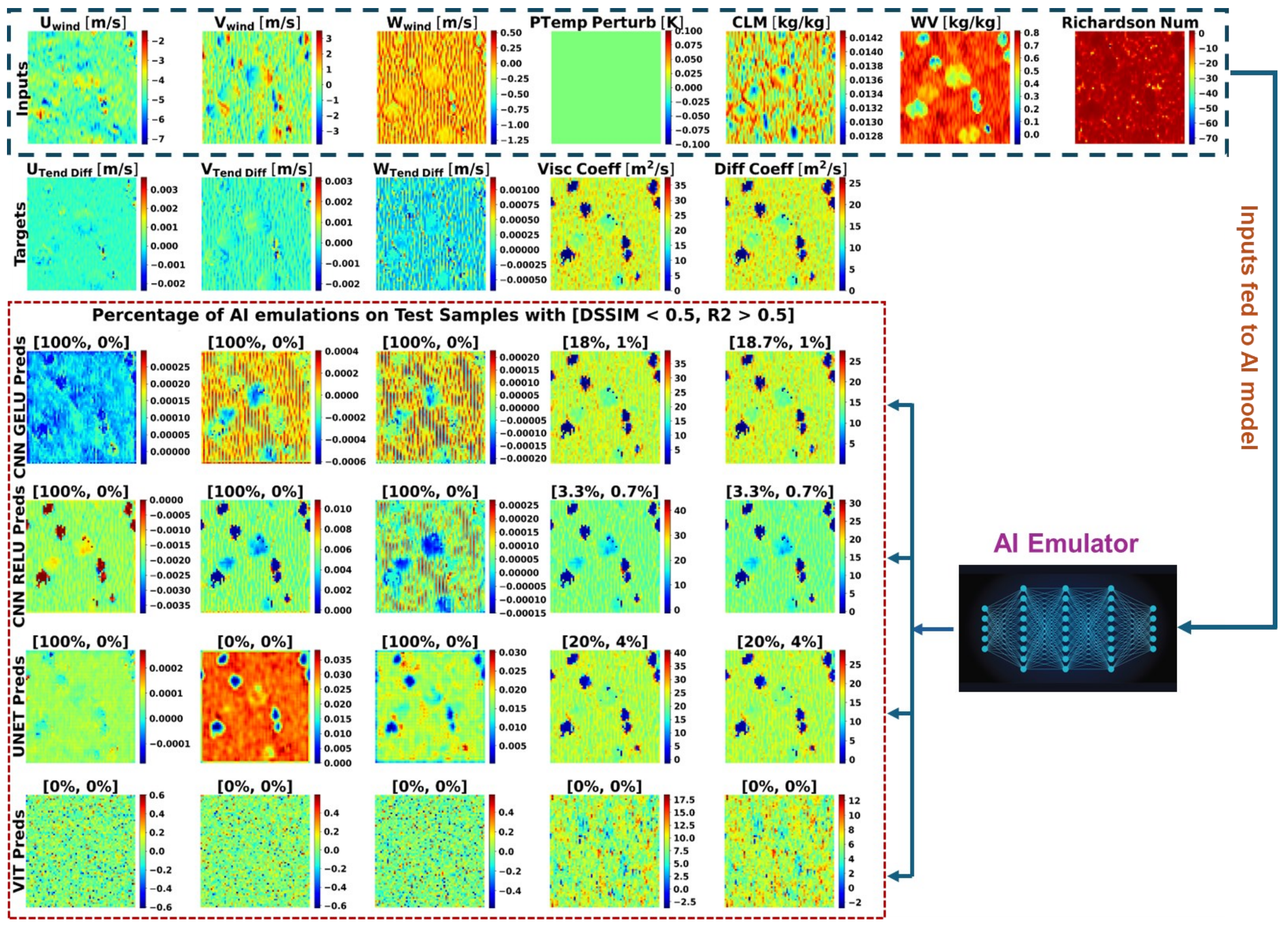


Fig 2: Representation of the emulation experiment, with input 7 channels on top row, 5 target variables in the 2nd row, and the AI predictions in the last 4 rows. Title of each plot in last 4 rows represent the percentage of test samples where the corresponding AI model has DSSIM < 0.5 and R2 > 0.5, which mean decent prediction accuracy.

Concluding remarks

- **Multi-variable prediction**: UNET and CNN (GELU/RELU) architectures demonstrated successful emulation with decent accuracy.
- **Accuracy (DSSIM < 0.5** and **R2 > 0.5**): Was limited to very few samples, indicating poor generalization and inefficient training.
- **Prediction variability**: Model Accuracy varied across the different target variables, as per the evaluation metrics used.
- Better models and more training data required for improving emulation performance.

Future Plans

- Expand training data (more **test cases**, full **3D data**), explore advanced AI techniques and evaluation metrics for model intercomparison.
- Test the **stability** of AI models by **integrating** them with MONC for real-time testing.
- Explore **XAI** methods to build trust and transparency in the AI emulation.

References

1. Pal, A. (2020). Deep learning emulation of subgrid-scale processes in turbulent shear flows. Geophysical Research Letters, 47(12), e2020GL087005.
2. Vinuesa, R., Brunton, S.L. Enhancing computational fluid dynamics with machine learning. Nat Comput Sci 2, 358–366 (2022).
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